

HERVE IRAKOZE NSUMBA

Bachelor of Science in Software Engineering

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PERSONAL PROFILE

Software Engineering student with hands-on experience building full-stack web applications, backend systems, and data pipelines. Experienced with Java, Spring Boot, Python, JavaScript, React, Next.js, PostgreSQL, Docker, and Git. Strong analytical and problem-solving skills developed through software engineering coursework, traineeships, and personal projects. Interested in backend engineering, data engineering, and technology-driven solutions for social and institutional impact.

EDUCATION

Bachelor of Science in Software Engineering

2023 – Dec 2027

Adventist University of Central Africa (AUCA), Kigali, Rwanda

Advanced Level — Physics, Chemistry, Biology (PCB)

Completed Jan 2023

Collège du Christ Roi, Nyanza, Rwanda

PROFESSIONAL EXPERIENCE

Software Engineer Trainee

Dec 2025 – May 2026

Africa to Silicon Valley (A2SV)

- Trained in data structures and algorithms with a focus on solving real-world competitive programming problems
- Applied algorithmic thinking to optimize application performance and build efficient, responsive solutions

React Developer Trainee

Nov 2025 – Jan 2026

Solvit Africa

- Built user interfaces using React with a component-based architecture
- Developed front-end applications and collaborated on responsive design implementation

PROJECTS

JobPortal Web Application

React, Spring Boot, Docker, REST API, PostgreSQL

- Built a full-stack application connecting job providers with job seekers, where employers post positions and candidates browse and apply
- Developed the REST API backend with Spring Boot and PostgreSQL, and the React frontend with responsive design
- Containerized the application using Docker for consistent development and deployment

Artha & Alita Apartments — Booking Platform

Next.js

- Collaboratively built a website and booking system for two apartments in Kigali, allowing users to browse listings, view details, and make reservations

- Implemented responsive UI and integrated booking workflows using Next.js

Electric Power Analytics — Data Pipeline

Python, PostgreSQL, GitHub Actions, Tableau

- Built an end-to-end data pipeline: World Bank API → PostgreSQL → cleaned analysis table → forecasts → Tableau dashboard
- Automated daily data refresh using GitHub Actions on free infrastructure with zero server maintenance
- Integrated an AI assistant that answers analytical questions in plain English

Idempotency Gateway (Pay-Once Protocol)

Spring Boot, REST API, PostgreSQL

- Developed a Spring Boot REST API that prevents duplicate payment processing using an Idempotency-Key header
- Simulated a fintech payment backend where repeated requests with the same key are processed exactly once

SKILLS & CERTIFICATIONS

Languages:	Java, Python, JavaScript, SQL
Frameworks:	Spring Boot, React, Next.js
Databases:	PostgreSQL, Snowflake
Tools:	Git & GitHub, Docker, Jupyter Notebook, Tableau
Other:	REST APIs, Data Analysis, Data Cleaning, Linux

Certifications: Object-Oriented Programming in Java — Coursera/IBM (2026) | Java Programming For Beginners — Coursera/IBM (2026) | Software Development Intermediate — Digital Talent Program, Rwanda ICT Chamber (2025) | Responsive Web Design — freeCodeCamp (2024) | Professional Scrum Master I — Scrum.org (2024)

INTERESTS & ACHIEVEMENTS

Passionate about backend software engineering, data engineering, cloud technologies, problem solving & algorithms, and leveraging technology for social and institutional impact.

Achievement: 50 Days Badge on LeetCode (2026) — consistent daily problem solving in data structures and algorithms.